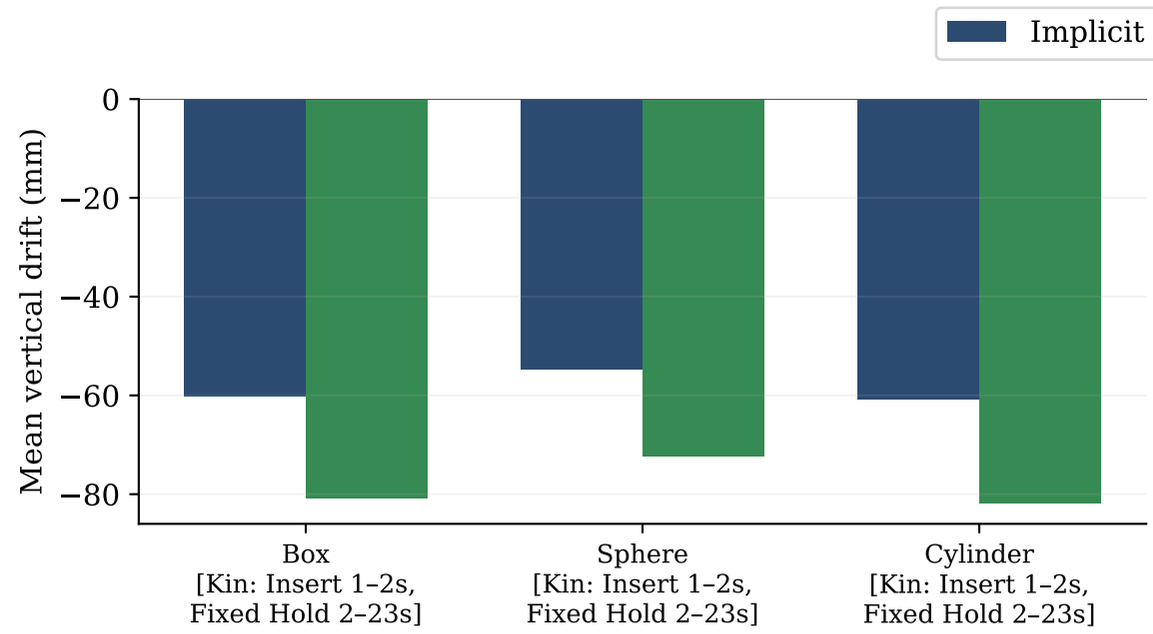
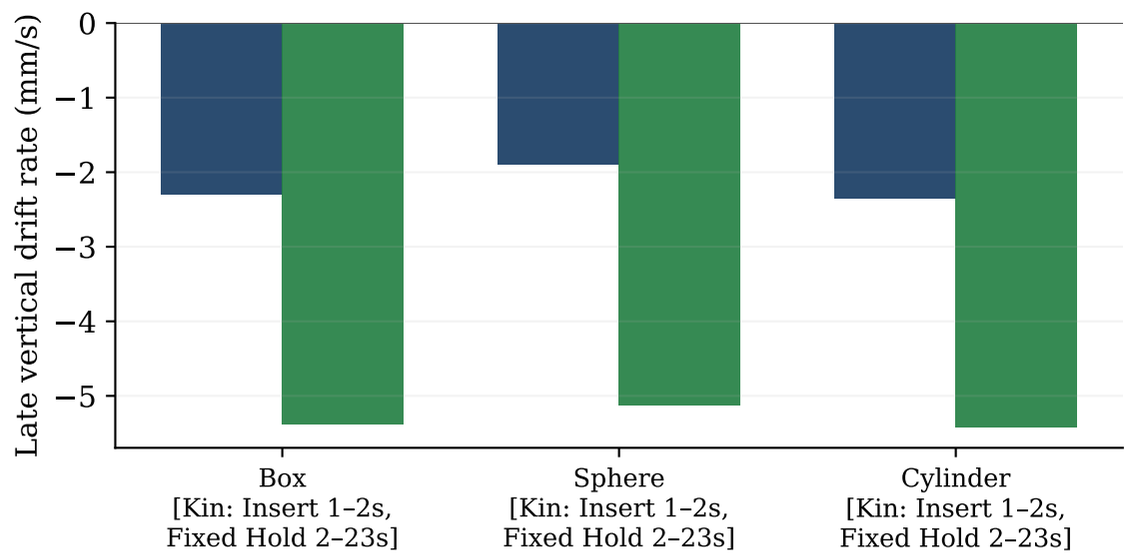
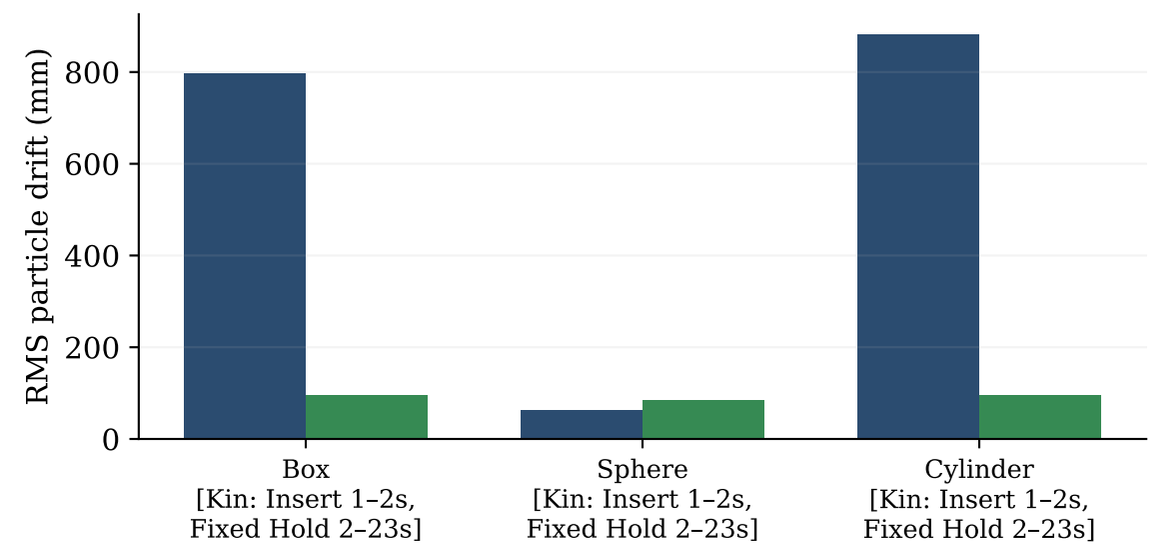
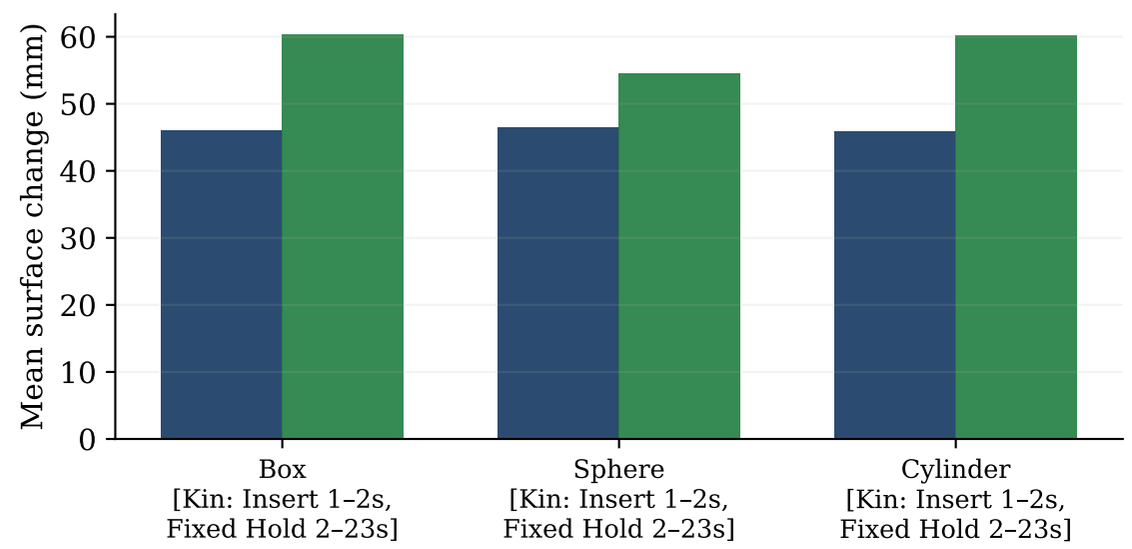
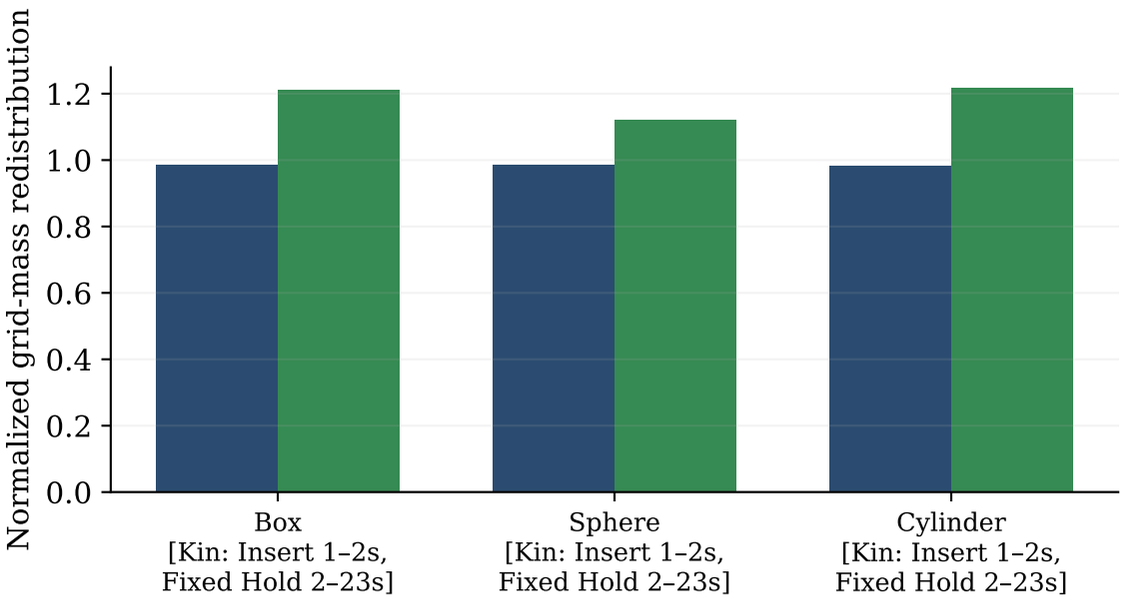
Kinematic Motion: Vertical Insertion (1–2 s) + Stationary Fixed-Pose Hold (2–23 s)











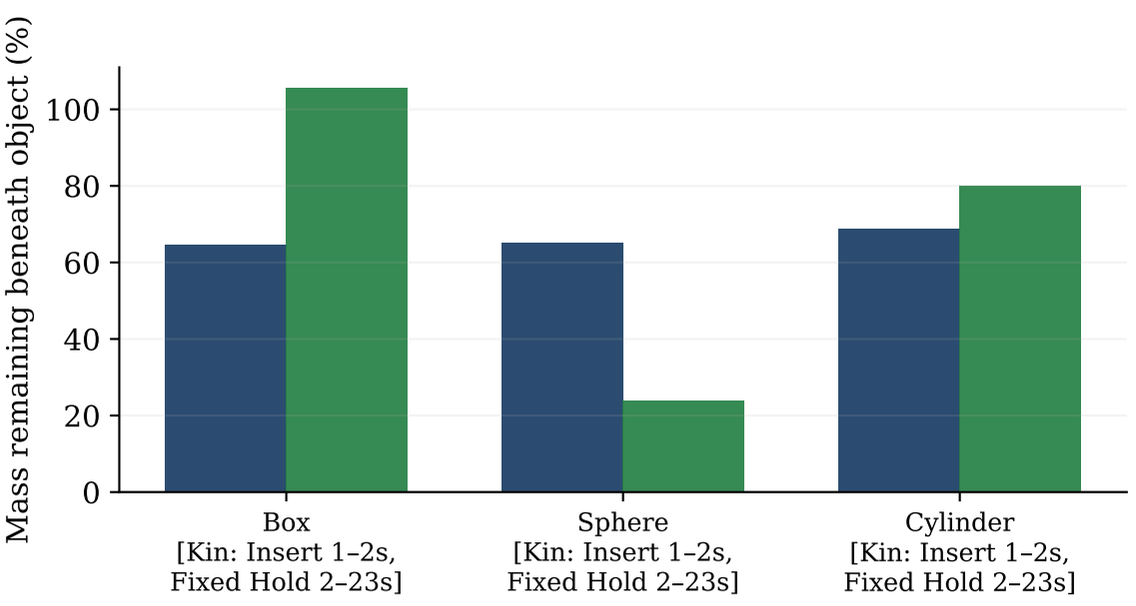


Figure 7: Frozen object-model checkpoint 10,800. Kinematic Motion: Object is inserted vertically (1–2 s) and held at an exactly fixed pose (2–23 s); measured relative to 3 s. Top row: mean change in world z at 23 s relative to 3 s, and late drift rate (13–23 s). Middle row: RMS particle drift and mean surface change. Bottom row: grid-mass redistribution and mass remaining beneath object.